

Chapter 4: Preventive Maintenance and Troubleshooting



IT Essentials v8.0

Chapter 4 - Sections & Objectives

- 4.1 Preventive Maintenance
  - Explain why preventive maintenance must be performed on personal computers.
  - Describe PC preventive maintenance.
- 4.2 Troubleshooting Process
  - Troubleshoot problems with PC and Peripheral devices
  - Describe each step of the troubleshooting process.
  - Identify common problems and solutions for PCs.
  - Troubleshoot computer components and peripherals using the six-step troubleshooting process.

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4.1 Preventive Maintenance

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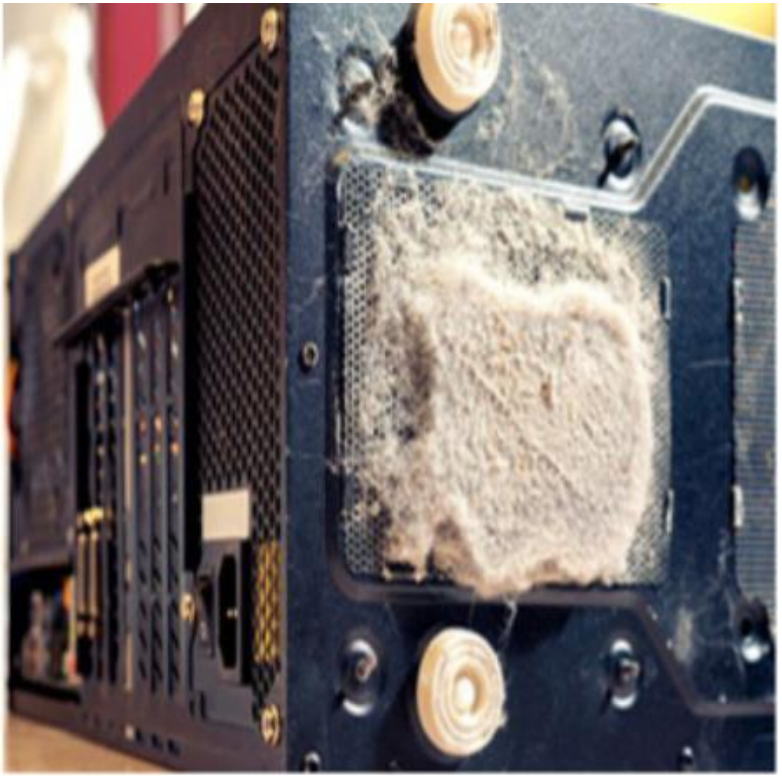
Benefits to Preventive Maintenance

- Preventive maintenance plans are developed based on at least two factors:
  - **Computer location or environment** - Dusty environments, such as construction sites, requires more attention than an office environment.
  - **Computer use** - High-traffic networks, such as a school network, might require additional scanning and removal of malicious software and unwanted files.

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Preventive Maintenance - Dust

- Use a cloth or a duster to clean the outside of the computer case.
- If using a cleaning product, put a small amount onto a cleaning cloth and then wipe the outside of the case.
- Dust on the outside of a computer can travel through cooling fans to the inside.
- Accumulated dust prevents the flow of air and reduces the cooling of components.
- Hot computer components are more likely to break down.
- Remove dust from the inside of a computer using a combination of compressed air, a low-air-flow ESD vacuum cleaner, and a small lint-free cloth.



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Preventive Maintenance – Internal Components

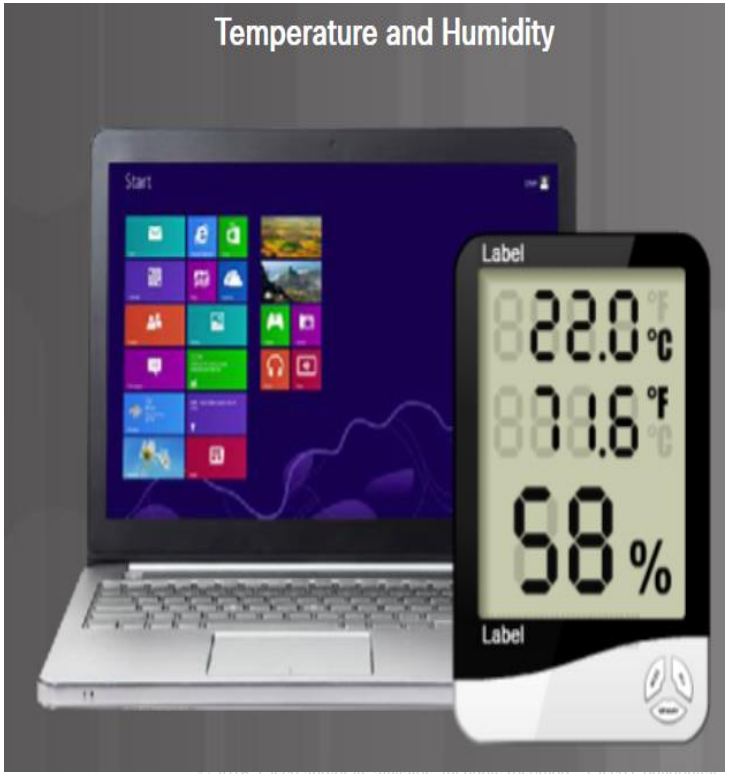
- A basic checklist of components to inspect for dust and damage includes:
  - **CPU heat sink and fan assembly**
  - **RAM modules**
  - **Storage devices**
  - **Adapter cards**
  - **Cables**
  - **Power devices**
  - **Keyboard and mouse**



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Preventive Maintenance – Environmental Concerns

- An optimal operating environment for a computer is clean, free of potential contaminants, and within the temperature and humidity range specified by the manufacturer.



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Preventive Maintenance – Software

- Verify that installed software is current.
  - Follow the policies of the organization when installing security updates, operating system, and program updates.
- Create a software maintenance schedule to:
  - Review and install the appropriate security, software, and driver updates.
  - Update the virus definition files and scan for viruses and spyware.
  - Remove unwanted or unused programs.
  - Scan hard drives for errors and defragment hard drives.

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4.2 Troubleshooting Process

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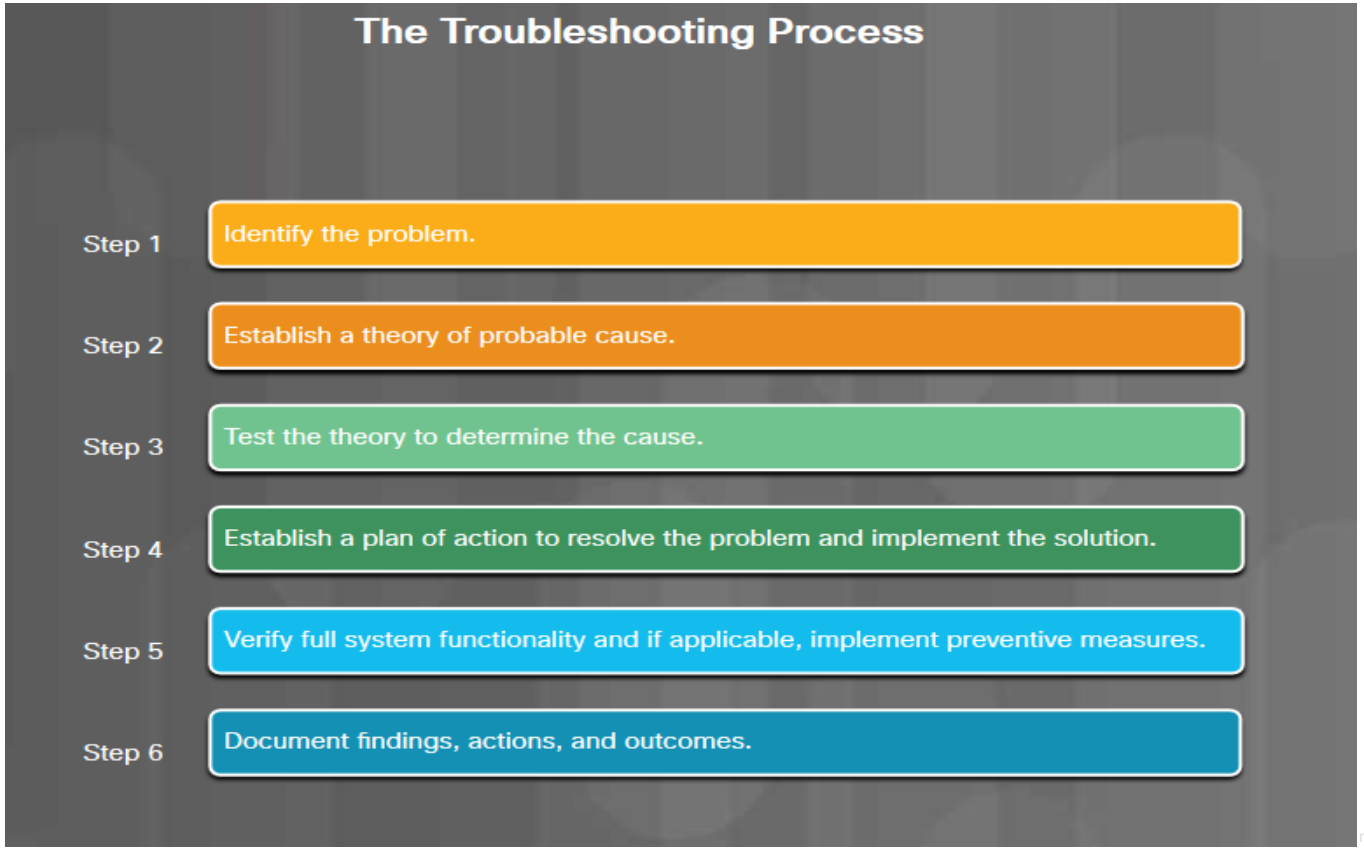
Introduction to Troubleshooting

- Troubleshooting requires an organized and logical approach to problems with computers and other components.
- Troubleshooting is a skill refined over time.
- Before you begin troubleshooting problems, always follow the necessary precautions to protect data on a computer.



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Troubleshooting Process Steps



Identify the Problem

Step 1: Identify the Problem.	
Customer Information	• Company Name • Contact Name • Address • Phone Number
Computer Configuration	• Manufacturer and Model • Operating System • Network Environment • Connection Type
Problem Description	• Open-ended Questions • Closed-ended Questions
Error Messages	
Beep Sequences	
LEDs	
POST	

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Establish a Theory of Probable Cause

Step 2. Establish a Theory of Probable Cause.

- Device is powered off.
- Power switch for an outlet is turned off.
- Surge protector is turned off.
- Loose external cable connections.
- Non-bootable disk in designated boot drive.
- Incorrect boot order in BIOS setup.

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Test the Theory to Determine the Cause

Step 3. Test the Theory to Determine the Cause.	
Common steps to determine cause	• Ensure the device is powered on. • Ensure the power switch for an outlet is turned on. • Ensure the surge protector is turned on. • Ensure external cable connections are secure. • Ensure that the designated boot drive is bootable. • Verify the boot order in BIOS setup.

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Establish a Plan of Action to Resolve the Problem and Implement the Solution

Step 4: Establish a Plan of Action to Resolve the Problem and Implement the Solution.	
If no solution is achieved in the previous step, further research is needed to implement the solution.	• Helpdesk repair logs • Other technicians • Manufacturer FAQ websites • Technical websites • News groups • Computer manuals • Device manuals • Online forums • Internet search

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Verify Full Functionality and, If Applicable, Implement Preventive Measures

Step 5: Verify Full System Functionality and if Applicable Implement Preventive Measures.

- Reboot the computer.
- Ensure multiple applications work properly.
- Verify network and Internet connections.
- Print a document from one application.
- Ensure all attached devices work properly.
- Ensure no error messages are received.

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Document Findings, Actions, and Outcomes

Step 6: Document Findings, Actions, and Outcomes

- Discuss the solution implemented with the customer.
- Have the customer verify that the problem has been solved.
- Provide the customer with all paperwork.
- Document the steps taken to solve the problem in the work order and in the technician's journal.
- Document any components used in the repair.
- Document the amount of time spent to resolve the problem.

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PC Common Problems and Solutions

- Computer problems can be attributed to hardware, software, networks, or some combination of the three. These are some common hardware problems:
- **Storage Device** - Storage device problems are often related to loose, or incorrect cable connections, incorrect drive and media formats, and incorrect jumper and BIOS settings.
- **Motherboard and Internal Components** - These problems are often caused by incorrect or loose cables, failed components, incorrect drivers, and corrupted updates.
- **Power Supply** - Power problems are often caused by a faulty power supply, loose connections, and inadequate wattage.
- **CPU and Memory** - Processor and memory problems are often caused by faulty installations, incorrect BIOS settings, inadequate cooling and ventilation, and compatibility issues.
- **Displays** – Display problems are often caused by incorrect settings, loose connections, and incorrect or corrupted drivers.

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Common Problems and Solutions for Storage Devices

Select each problem

- The computer does not recognize a storage device.
- The computer does not recognize an optical disc.
- The computer will not eject the optical disc.
- The computer does not recognize a removable, external drive.
- A media reader cannot read a memory card that works properly.
- Retrieving or saving data from the USB flash drive is slow.

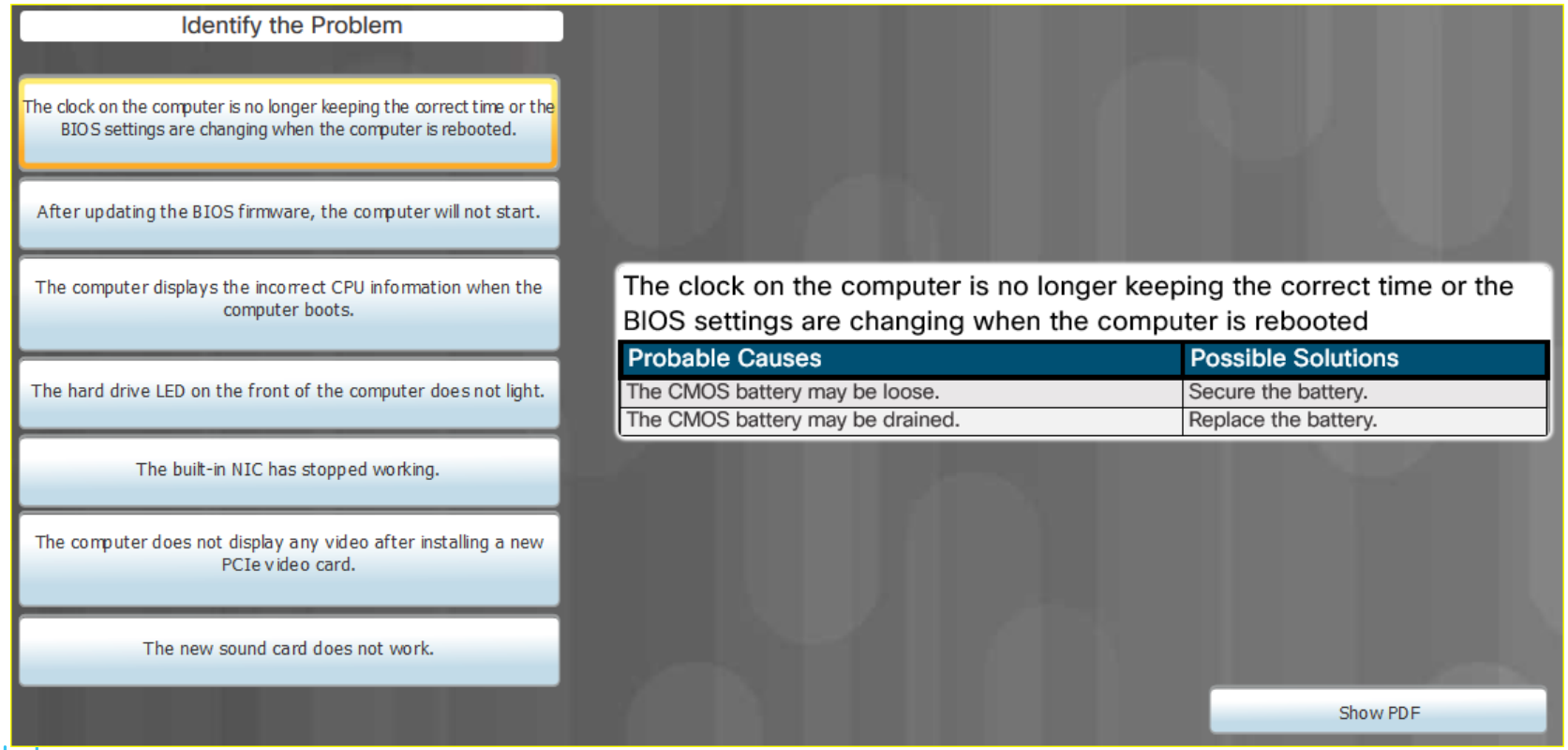
The computer does not recognize a storage device

Probable Causes	Possible Solutions
The power cable is loose.	Secure the power cable.
The data cable is loose.	Secure the data cable.
The jumpers are set incorrectly.	Reset the jumpers.
A storage device failed	Replace the storage device.
The storage device setting in BIOS are incorrect.	Reset the storage device settings in BIOS.

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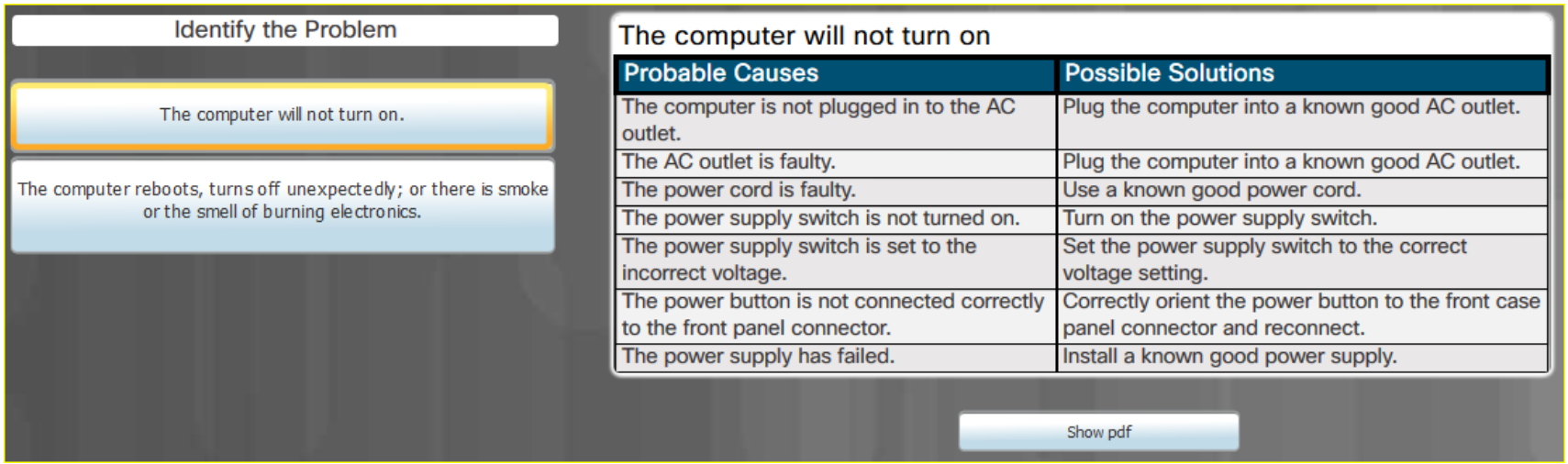
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Common Problems and Solutions for Motherboards and Internal Components



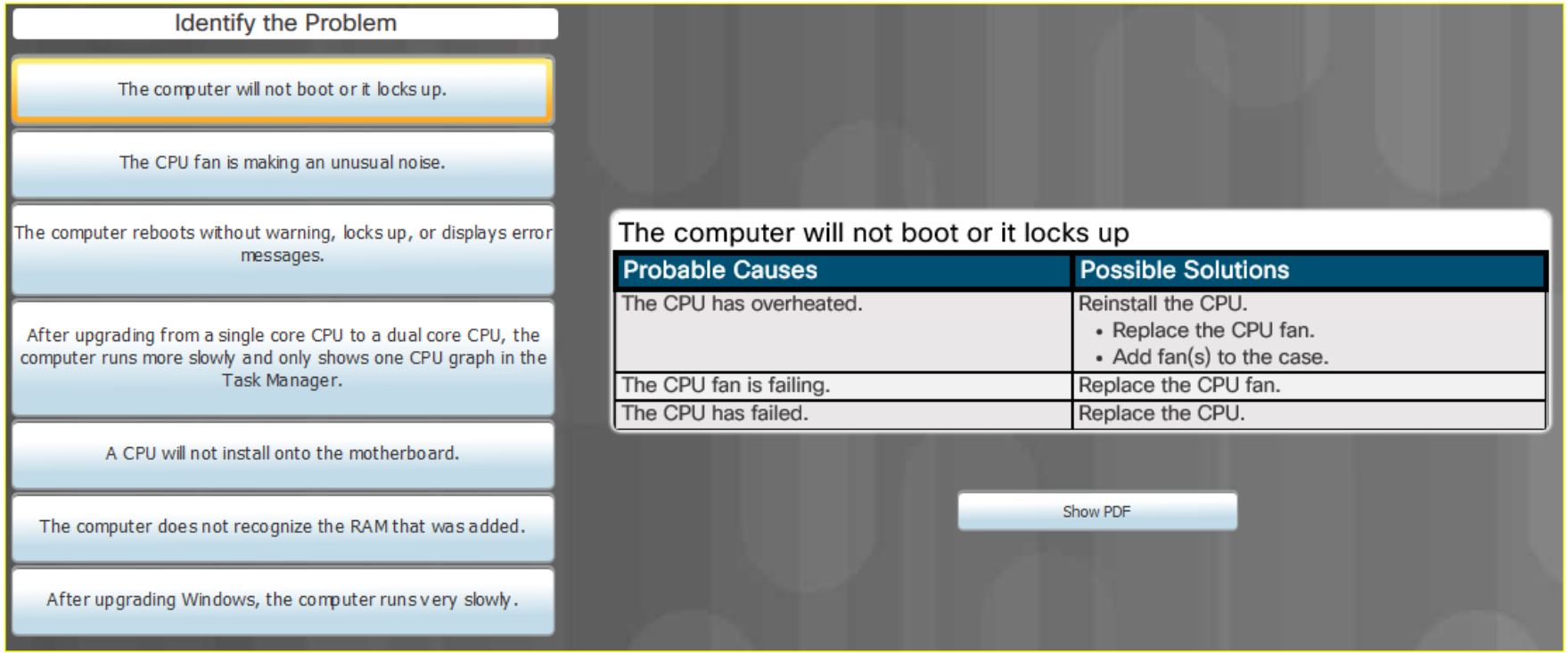
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Common Problems and Solutions for Power Supplies



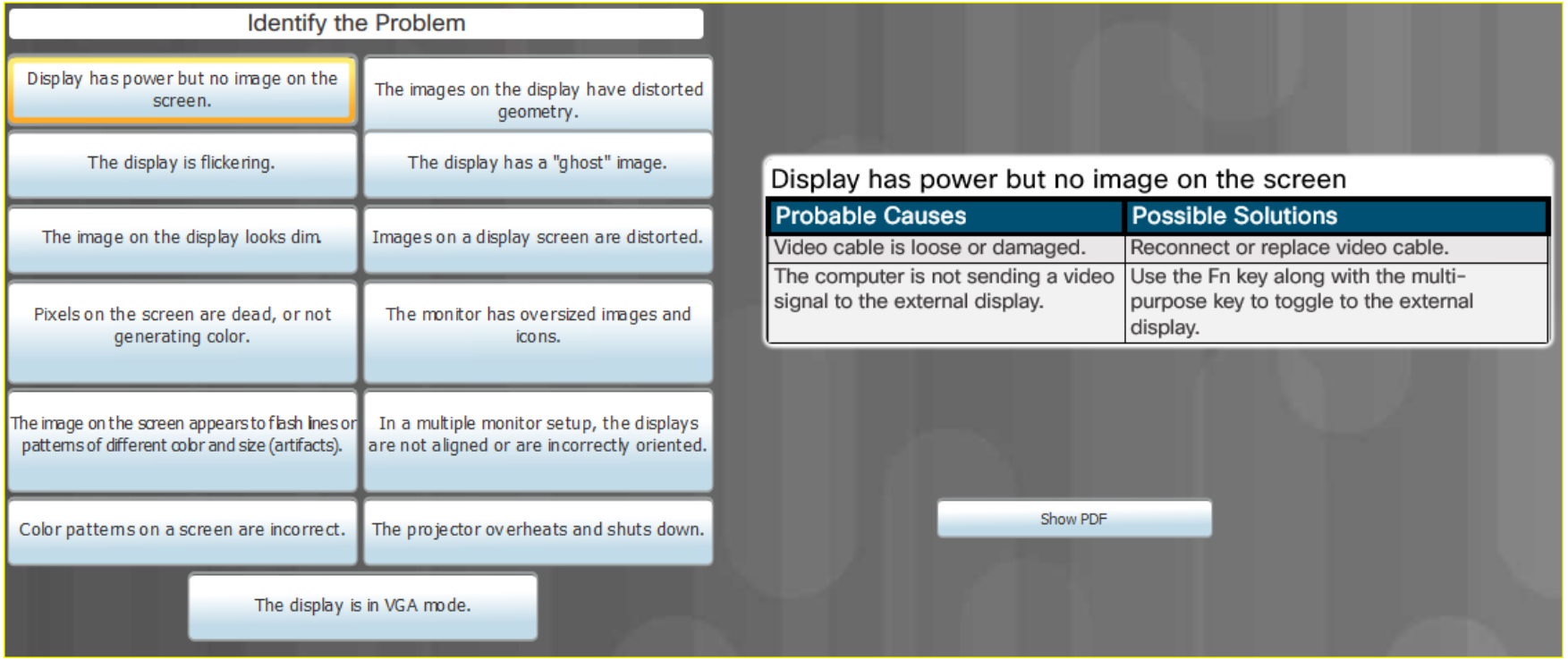
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Common Problems and Solutions for CPUs and Memory



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Common Problems and Solutions for Displays



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Personal Reference Tools

- Personal reference tools include troubleshooting guides, manufacturer manuals, quick reference guides, and repair journals. In addition to an invoice, a technician keeps a journal of upgrades and repairs:
- **Notes** - Make notes as you go through the troubleshooting and repair process. Refer to these notes to avoid repeating steps and to determine what needs to be done next.
- **Journal** - Include descriptions of the problem, possible solutions that have been tried to correct the problem, and the steps taken to repair the problem. Note any configuration changes made to the equipment and any replacement parts used in the repair. Your journal, along with your notes, can be valuable when you encounter similar situations in the future.
- **History of repairs** - Make a detailed list of problems and repairs, including the date, replacement parts, and customer information. The history allows a technician to determine what work has been performed on a specific computer in the past.

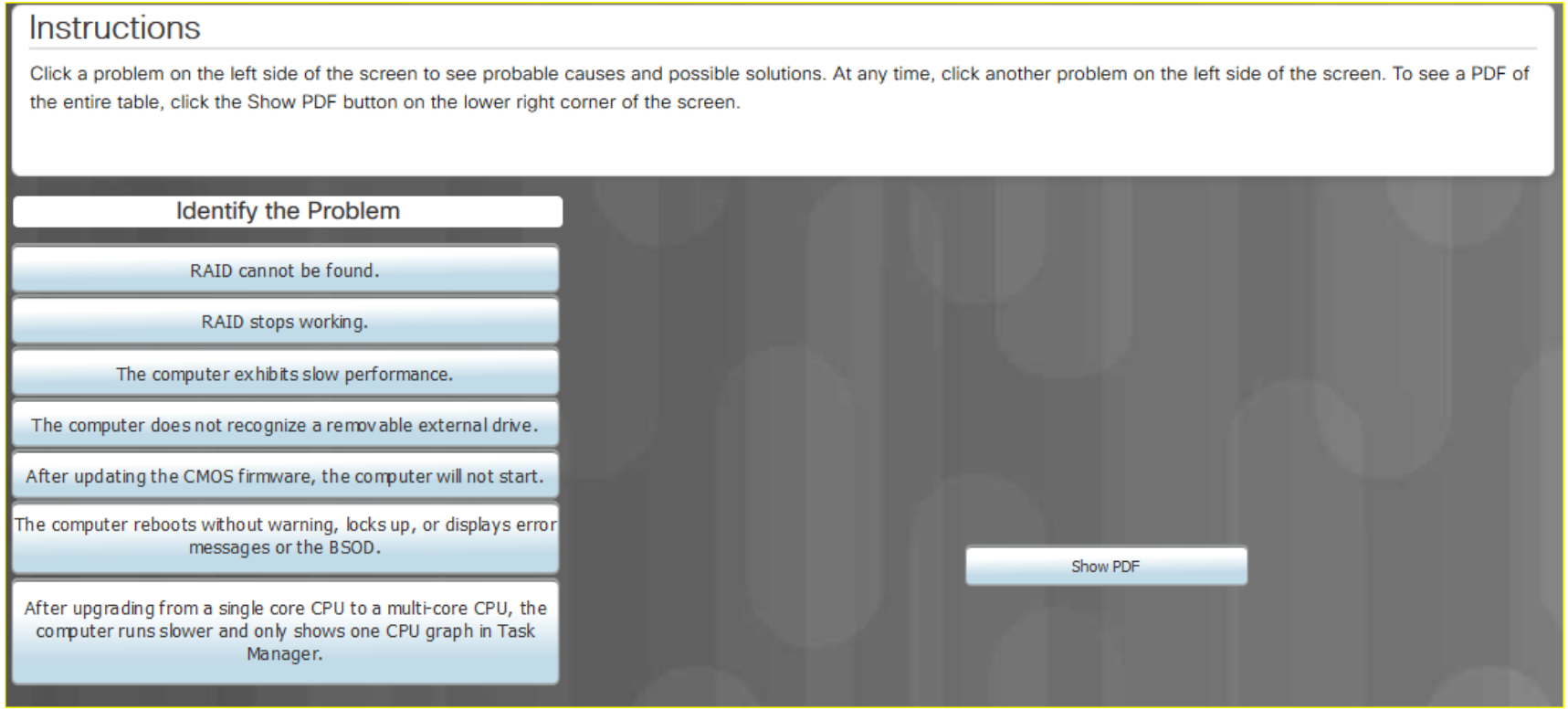
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Internet Reference Tools

- The Internet is an excellent source of information about specific hardware problems and possible solutions:
  - Internet search engines
  - News groups
  - Manufacturer FAQs
  - Online computer manuals
  - Online forums and chat
  - Technical websites

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Advanced Problems and Solutions for Hardware



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Lab – Using a Multimeter and a Power Supply Tester

- In this lab, you will learn how to use and handle a multimeter and a power supply tester.

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Lab – Troubleshoot Hardware Problems

- In this lab, you will diagnose the cause of various hardware problems and solve them.

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4.3 Chapter Summary

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- Explain why preventive maintenance must be performed on personal computers.
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